Course: BTech **Semester:** 7

Prerequisite: Basics of Python & Machine Learning

Rationale : Natural language processing (NLP) is a crucial part of artificial intelligence (AI). This course will introduce about the core concepts related to Natural Language Processing and its applications. A broad introduction about incorporation of Deep Learning techniques in Natural Language Processing is considered.

Teaching and Examination Scheme

Teaching Scheme					Examination Scheme					Total
					Internal Marks			External Marks		
Lecture Hrs/Week	Tutorial Hrs/Week	Lab Hrs/Week	Seminar Hrs/Week	Credit	T	CE	P	T	P	
0	0	2	0	1	-	-	20	-	30	50

SEE - Semester End Examination, **T** - Theory, **P** - Practical

Course Outcome

After Learning the Course the students shall be able to:

1. Understand the concepts of Natural Language Processing.
2. Understand the concepts of Deep Learning
3. Apply several models of Deep Learning in Natural Language Processing
4. Analyze the importance of Recurrent Neural Network & LSTM in NLP.
5. Understand about recent developments in the field of Chatbots
6. Classify and Implement Sentimental Analysis techniques

List of Practical	
1.	**Implementation of preprocessing of Text with NLTK (Tokenization, Stemming, Lemmatization and removal of stop words in NLP** Implementation of preprocessing of Text with NLTK (Tokenization, Stemming, Lemmatization and removal of stop words in NLP
2.	**Implementation to Convert the text to word count vectors with ScikitLearn (CountVectorizer).** Implementation to Convert the text to word count vectors with ScikitLearn (CountVectorizer).
3.	**Implementation to Convert the text to word frequency vectors with ScikitLearn (TfidfVectorizer).** Implementation to Convert the text to word frequency vectors with ScikitLearn (TfidfVectorizer).
4.	**Implementation to Convert the text to unique integers with ScikitLearn (HashingVectorizer).** Implementation to Convert the text to unique integers with ScikitLearn (HashingVectorizer).
5.	**Use the Keras deep learning library and split words with (text_to_word_sequence).** Use the Keras deep learning library and split words with (text_to_word_sequence).
6.	**Use the Keras deep learning library and write a code for encoding with (one_hot).** Use the Keras deep learning library and write a code for encoding with (one_hot).
7.	**Use the Keras deep learning library and write a code for Hash Encoding with (hashing_trick).** Use the Keras deep learning library and write a code for Hash Encoding with (hashing_trick).
8.	**Use the Keras deep learning library give a demo of Tokenizer API.** Use the Keras deep learning library give a demo of Tokenizer API.
9.	**Perform an experiment to do sentimental analysis on any of the dataset (like twitter tweets, movie review etc.).** Perform an experiment to do sentimental analysis on any of the dataset (like twitter tweets, movie review etc.).
10.	**Perform an experiment using Gensim python library for Word2Vec Embedding.** Perform an experiment using Gensim python library for Word2Vec Embedding.